GÖDEL–DUMMETT LOGIC OVER A FINITE CATEGORY OF MODELS: THE MODELS OF THE TRANSLATED THEORY ARE STRINGS OF ARROWS, AND A SUBGROUP OBSTRUCTION TO COMMUTING WITH THE IMAGINARIES

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Abstract. Let $\mathcal{X}$ be a finite category in which every endomorphism is invertible, and $\mathcal{P}$ the pretopos of finite set-valued functors on $\mathcal{X}$, a first-order theory whose models are the objects of $\mathcal{X}$. For an intermediate logic L proving the linearity axiom $(p \to q) \vee (q \to p)$ we describe the theory read in first-order logic over L: its models are the strings of non-invertible arrows of $\mathcal{X}$, of length bounded by the depth of L, and its pretopos completion is the category of finite set-valued functors on the category $\mathcal{S}$ of strings, whose arrows are the arrows of $\mathcal{X}$ induced by isomorphisms of final segments of the domain onto the codomain. The translation into L commutes with the imaginaries if and only if every finite functor on $\mathcal{S}$ embeds in one pulled back from $\mathcal{X}$, a decidable condition. It holds for classical logic and the logic of the two-element chain. For the other logics it holds for finite posets, for products of a finite poset and a finite group, and when $\mathcal{X}$ has only identity automorphisms; it fails when $\langle H \cap S_1, H \cap S_2\rangle \neq H \cap \langle S_1, S_2\rangle$ for the group H of the automorphisms of the first object of a string which extend to the string and the stabilisers S_1, S_2 of two arrows of $\mathcal{S}$ into it. In a family of categories with three levels it holds exactly when a subgroup A of an abelian group distributes over subgroups T_j, $A \cap \sum_J T_j = \sum_J (A \cap T_j)$. The main tool is the linearity of the canonical Kripke model with elements: two homomorphisms out of a model of the translated theory are comparable.

1. Introduction

For an intermediate logic L and a small coherent category $\mathcal{C}$ let $\mathsf{H}_L\mathcal{C}$ be the syntactic category of the internal coherent theory of $\mathcal{C}$ in first-order logic over L, and P the pretopos completion, which adds the imaginaries. There is a full and faithful comparison

$$c^L_{\mathcal{C}} \colon \mathsf{H}_L\mathsf{P}\mathcal{C} \longrightarrow \mathsf{P}\mathsf{H}_L\mathcal{C}$$

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[6]. For intuitionistic logic it is an equivalence only if $\mathcal{C}$ is Boolean [5]; for the logics proving the linearity axiom it is an equivalence for all propositional theories [6], and for the first-order theory of pure equality it is an equivalence only for classical logic [6]. This paper studies the logics L proving linearity over the first-order theories which are closest to the propositional ones: those with finitely many models up to isomorphism, all sorts being finite in each of them.

Let $\mathcal{X}$ be a finite category and $\mathcal{P} = \mathbf{FinSet}^{\mathcal{X}}$ the pretopos of functors from $\mathcal{X}$ to finite sets; if idempotents split in $\mathcal{X}$, the models of $\mathcal{P}$ are the objects of $\mathcal{X}$ and the homomorphisms are the arrows. For a finite poset $\mathcal{X}$ this is a finite propositional theory. We assume that $\mathcal{X}$ is an *EI-category*: every endomorphism is an isomorphism. A *string* is a sequence $s = (m_0 \xrightarrow{a_1} m_1 \to \ldots \xrightarrow{a_r} m_r)$ of non-invertible arrows, $r \geq 0$, and $\mathcal{S}_n$ is the category of the strings with at most n objects, one in each isomorphism class, an arrow $s \to s'$ being an arrow $m_0 \to m'_0$ of $\mathcal{X}$ of the form $\iota_0 a_j \cdots a_1$ for an isomorphism of strings ι from the final segment $(m_j \to \cdots \to m_r)$ onto s'. Let $\pi \colon \mathcal{S}_n \to \mathcal{X}$ be the first object.

Theorem 1.1. *Let $\mathcal{X}$ be a finite EI-category, $\mathcal{P} = \mathbf{FinSet}^{\mathcal{X}}$, and L an intermediate logic proving the linearity axiom, with $n = n_L \in \{1, 2, \ldots, \infty\}$ the supremum of the lengths of the finite chains which are frames of L. Then*

$$\mathsf{PH}_L\mathcal{P} \;\simeq\; \mathbf{FinSet}^{\mathcal{S}_n},$$

the unit of H_L followed by that of P corresponding to $\pi^* \colon \mathbf{FinSet}^{\mathcal{X}} \to \mathbf{FinSet}^{\mathcal{S}_n}$, and $\mathsf{H}_L\mathcal{P}$ to the full subcategory of the subfunctors of the functors $\pi^* A$. Consequently $c^L_{\mathcal{P}}$ is an equivalence if and only if every functor $W \colon \mathcal{S}_n \to \mathbf{FinSet}$ is a subfunctor of $\pi^* Y$ for some $Y \colon \mathcal{X} \to \mathbf{FinSet}$, if and only if the unit $W \to \pi^* \mathrm{Lan}_\pi W$ of the left Kan extension is injective for every W.

For $n = 1$, classical logic, $\mathcal{S}_1$ is a skeleton of the groupoid of the isomorphisms of $\mathcal{X}$. The condition of the theorem is combinatorial. We prove:

- it holds if for all arrows $v \colon t' \to t$, $u \colon t' \to s$ of $\mathcal{S}_n$ every arrow g of $\mathcal{X}$ with $gv = u$ is an arrow $t \to s$ of $\mathcal{S}_n$ (Proposition 6.1); this is the case for finite posets, which gives again the theorem of [6] for finite propositional theories, for the product of a finite poset and a finite group (Corollary 6.2), and for all finite EI-categories if $n_L \leq 2$ (Corollary 6.3);
- more generally it holds if, for every string s, the arrows of $\mathcal{S}_n$ into s are a natural retract of the arrows of $\mathcal{X}$ which are connected to them (Proposition 6.4); this is the case if $\mathcal{X}$ has no automorphisms other than identities (Corollary 6.5);
- it fails if for a string t, with $H \leq G = \mathrm{Aut}_{\mathcal{X}}(m_0)$ the group of the first components of its automorphisms, and two arrows of $\mathcal{S}_n$ into t with stabilisers $S_1, S_2 \leq G$ one has $\langle H \cap S_1, H \cap S_2 \rangle \neq H \cap \langle S_1, S_2 \rangle$

(Proposition 6.7); this happens in a category with four objects and fifteen arrows (Example 6.8), for every L with $n_L \geq 3$;
- it fails if an arrow g of $\mathcal{X}$ which is not an arrow of $\mathcal{S}_n$ becomes one after composition with two arrows v_1, v_2 of $\mathcal{S}_n$, in two ways which are not connected inside $\mathcal{S}_n$ (Proposition 6.9); a category with seventeen arrows shows that this is not a consequence of the subgroup obstruction (Example 6.10);
- for the categories with objects $0_1, \ldots, 0_m, 1, 2$ given by subgroups $A, T_1, \ldots, T_m$ of a finite abelian group it holds, for $n_L \geq 3$, if and only if $A \cap \sum_{j\in J} T_j = \sum_{j\in J}(A \cap T_j)$ for every set J of indices (Proposition 6.11); among them are a category with 28 arrows for which the comparison is an equivalence although there is no retraction, and, for every $m \geq 3$, a category for which it is not an equivalence although it is one for every full subcategory omitting one of the objects 0_j, and the obstruction of Proposition 6.9 is absent (Example 6.12);
- in general the condition has to be tested on finitely many functors only, one for each string, so that it is decidable (Proposition 6.13).

For classical logic the assumption on the endomorphisms is not needed: the comparison is an equivalence for every finite category in which idempotents split (Proposition 5.3). The results are summarised as follows, $\mathcal{X}$ being a finite category in which idempotents split.

logic L	$\mathcal{X}$	$c^L_{\mathcal{P}}$ is an equivalence
classical ($n_L = 1$)	any	always
linear, $n_L = 2$	EI	always
linear, $3 \leq n_L \leq \infty$	EI	iff the units $W \to \pi^*\mathrm{Lan}_\pi W$ are injective; not always
intuitionistic	any	only for groupoids [5]

So, in contrast with the propositional case, the Gödel–Dummett translation does not commute with the imaginaries of all these theories, the obstructions are of a group-theoretic nature, and they appear exactly from the three-element chain on. For intuitionistic logic the comparison is an equivalence only if $\mathcal{X}$ is a groupoid [5, Thm. 1.1]; the product of the poset $0 < 1$ and the group $\mathbb{Z}/2$, for which an object outside the image is given in [5, Ex. 7.16], is among the categories for which the comparison is an equivalence for all L proving linearity.

What is new and what is known. New, to the author's knowledge: Theorem 1.1; the linearity of the canonical model with elements (Lemma 3.1) and the description of the models by strings (Proposition 4.2); the classical case (Proposition 5.3); Propositions 6.1, 6.4, 6.7, 6.9, 6.11 and 6.13 with their corollaries, and Examples 6.6, 6.8, 6.10 and 6.12. Known and used: the translation H_L, the comparison and the frames of the logics proving linearity [6]; the completeness theorem of coherent logic and the pretopos completion [4]; Joyal's completeness theorem for Kripke models in the form of a truth lemma for the category of coherent models, proved by the method

of diagrams as in [5, Lemma 7.13]; and the fact that the prime filters above a prime filter of a linear Heyting algebra form a chain [3], of which Lemma 3.1 is the version with elements. The description of the models by chains in the propositional case, where $\mathcal{X}$ is a finite poset, is known [1, 2, 6]. EI-categories are standard in the representation theory of finite categories; we use only the definition.

2. The setting

Conventions are those of [6, 5]. Let $\mathcal{X}$ be a finite category in which idempotents split, $\mathcal{P} = \mathbf{FinSet}^{\mathcal{X}}$, $y(x) = \mathcal{X}(x, -)$, and $y(\gamma)\colon y(x') \to y(x)$ the composition with $\gamma\colon x \to x'$. Every A in $\mathcal{P}$ is covered by the morphisms $q_a\colon y(x) \to A$, $a \in A(x)$, $q_a(\varphi) = A(\varphi)(a)$. The models of $\mathcal{P}$ are, up to isomorphism, the evaluations at the objects of $\mathcal{X}$, and the homomorphisms between these are the arrows of $\mathcal{X}$ [5, §7].

Let L be an intermediate logic and $\mathsf{H}_L\mathcal{P}$ the syntactic category of the internal theory of $\mathcal{P}$ in first-order logic over L [6, §2]: intuitionistic first-order logic with the additional axioms $\lambda(\varphi_1, \dots, \varphi_k)$, for $\lambda(p_1, \dots, p_k) \in L$ and first-order formulas φ_i in a common context, so that the formulas of L are valid in every lattice of subobjects of $\mathsf{H}_L\mathcal{P}$. Its objects are the subobjects $U \leq \eta A$, A in $\mathcal{P}$, and we omit η. The subobjects of the objects A of $\mathcal{P}$ in $\mathsf{H}_L\mathcal{P}$ are obtained from those in $\mathcal{P}$ by finite meets and joins, $\Rightarrow$, and f^*, $\exists_f$, $\forall_f$ for the morphisms f of $\mathcal{P}$. Since $\mathcal{P}$ is a pretopos, $\mathsf{H}_L\mathsf{P}\mathcal{P} \simeq \mathsf{H}_L\mathcal{P}$, and $c^L_{\mathcal{P}}$ is the unit $\mathsf{H}_L\mathcal{P} \to \mathsf{P}\mathsf{H}_L\mathcal{P}$; it is an equivalence if and only if $\mathsf{H}_L\mathcal{P}$ is a pretopos [6, Lemma 2.2].

A *model* of $\mathsf{H}_L\mathcal{P}$ is a coherent functor $\hat{G}\colon \mathsf{H}_L\mathcal{P} \to \mathbf{Set}$; it is not required to be a Heyting functor. Its restriction to $\mathcal{P}$ is isomorphic to the evaluation at an object m; we fix such an isomorphism and regard $\hat{G}(U)$, for $U \leq A$, as a subset of $A(m)$, and say that $\hat{G}$ is *over* m. A natural transformation $\hat{G} \to \hat{G}'$ of models over m, m' restricts to an arrow $\varphi\colon m \to m'$ and is determined by it; so a *homomorphism* $\hat{G} \to \hat{G}'$ is an arrow $\varphi\colon m \to m'$ of $\mathcal{X}$ with $A(\varphi)(\hat{G}(U)) \subseteq \hat{G}'(U)$ for all $U \leq A$. Let $\mathcal{K}$ be the category of the models and homomorphisms, with one model in each isomorphism class; the functor $\mathcal{K} \to \mathcal{X}$ is faithful.

Lemma 2.1. *An arrow $\varphi\colon m \to m'$ is a homomorphism $\hat{G} \to \hat{G}'$ if and only if $\varphi \in \hat{G}'(U)$ for every $U \leq y(m)$ with $\mathrm{id}_m \in \hat{G}(U)$.*

Proof. If $b \in \hat{G}(U)$, $U \leq A$, then $b = q_b(\mathrm{id}_m)$, so $\mathrm{id}_m \in \hat{G}(q_b^*U)$, and $A(\varphi)(b) = q_b(\varphi)$; so $A(\varphi)(b) \in \hat{G}'(U)$ iff $\varphi \in \hat{G}'(q_b^*U)$. □

Lemma 2.2 (truth lemma)**.** *Let $\hat{G}$ be a model over m, $f\colon B \to A$ a morphism of $\mathcal{P}$, $U, V \leq B$ in $\mathsf{H}_L\mathcal{P}$, $\alpha \in A(m)$ and $\beta \in B(m)$.*

(a) *$\alpha \in \hat{G}(\forall_f U)$ if and only if for every homomorphism $\varphi\colon \hat{G} \to \hat{G}'$ and every $\beta' \in B(m')$ with $f(\beta') = A(\varphi)(\alpha)$ one has $\beta' \in \hat{G}'(U)$.*

(b) $\beta \in \hat{G}(U \Rightarrow V)$ *if and only if for every homomorphism* $\varphi\colon \hat{G} \to \hat{G}'$, $B(\varphi)(\beta) \in \hat{G}'(U)$ *implies* $B(\varphi)(\beta) \in \hat{G}'(V)$.

Proof. The implications from left to right hold because φ preserves $\forall_f U$ and $U \Rightarrow V$, and $f^*\forall_f U \leq U$, $(U \Rightarrow V) \wedge U \leq V$. For the converse of (a) let $\alpha \notin \hat{G}(\forall_f U)$, and consider the classical theory of the coherent functors on $\mathsf{H}_L\mathcal{P}$ with a constant c_b for every element b of the sets $\hat{G}(X)$, X an object of $\mathsf{H}_L\mathcal{P}$, the axioms $g(c_b) = c_{g(b)}$ for the morphisms g of $\mathsf{H}_L\mathcal{P}$, a new constant d of sort B and the axioms $f(d) = c_\alpha$ and $\neg U(d)$. If it had no model, a finite part, written as a subobject θ of a product of A with finitely many objects, with a point in $\hat{G}(\theta)$ over α, would give $V_0 := \exists_{\pi_0}\theta \leq A$ with $\alpha \in \hat{G}(V_0)$ and $f^*V_0 \leq U$ by the completeness theorem for the coherent category $\mathsf{H}_L\mathcal{P}$; then $V_0 \leq \forall_f U$, contradicting $\alpha \notin \hat{G}(\forall_f U)$. A model of the theory is a coherent functor $\hat{G}'$ with a natural transformation $\hat{G} \to \hat{G}'$ and an element $\beta' \in B(m') \setminus \hat{G}'(U)$ over the image of α. This is the argument of [5, Lemma 7.13]. (b) is the same, with the axioms $U(c_\beta)$ and $\neg V(c_\beta)$. □

So the evaluation $U \mapsto (\hat{G} \mapsto \hat{G}(U))$ is a Heyting functor $\mathsf{H}_L\mathcal{P} \to \mathbf{Set}^{\mathcal{K}}$, which is conservative by the completeness theorem of coherent logic: if $\hat{G}(U) \subseteq \hat{G}(V)$ for all models, then $U \leq V$.

Kripke models. Let K be a finite poset such that every $\uparrow k$ is a frame of L, and $M\colon K \to \mathcal{X}$ a functor. Then $\mathbf{Set}^K$ is an L-category, and M extends to a Heyting functor $\mathsf{H}_L\mathcal{P} \to \mathbf{Set}^K$ [6, Lemma 2.3]; its value at $U \leq A$ is the family of the subsets of $A(M_k)$ defined by the forcing relation of M. The evaluation at a node k is a model of $\mathsf{H}_L\mathcal{P}$ over M_k, and the transitions $M(k \leq l)$ are homomorphisms between these models.

If L proves the linearity axiom, the finite frames of L are the finite posets in which every $\uparrow k$ is a chain with at most n_L elements [6, Lemma 3.1].

3. Two homomorphisms out of a model are comparable

Lemma 3.1 (linearity)**.** *Let $\mathcal{X}$ be a finite category in which idempotents split, L an intermediate logic proving $(p \to q) \vee (q \to p)$, and $\varphi\colon \hat{G} \to \hat{G}'$, $\psi\colon \hat{G} \to \hat{G}''$ homomorphisms of models of $\mathsf{H}_L\mathcal{P}$. Then there is a homomorphism $\chi\colon \hat{G}' \to \hat{G}''$ with $\chi\varphi = \psi$, or a homomorphism $\chi'\colon \hat{G}'' \to \hat{G}'$ with $\chi'\psi = \varphi$.*

Proof. Let the models be over m, m', m'', and suppose that there is no such χ. The set of the arrows $\chi\colon m' \to m''$ of $\mathcal{X}$ with $\chi\varphi = \psi$ is finite, and none of them is a homomorphism; by Lemma 2.1 there is for each of them $U_\chi \leq y(m')$ with $\mathrm{id}_{m'} \in \hat{G}'(U_\chi)$ and $\chi \notin \hat{G}''(U_\chi)$. Let U' be the meet of the U_χ (the top element, if there is no such arrow), and $E' := \exists_{y(\varphi)}U' \leq y(m)$. Then $\varphi \in \hat{G}'(E')$, since $\varphi = y(\varphi)(\mathrm{id}_{m'})$ and $\mathrm{id}_{m'} \in \hat{G}'(U')$; and $\psi \notin \hat{G}''(E')$, since an element of $\hat{G}''(U')$ over ψ is an arrow χ with $\chi\varphi = \psi$ and $\chi \in \hat{G}''(U')$. If there is no χ' either, we get in the same way $E'' \leq y(m)$ with

$\psi \in \hat{G}''(E'')$ and $\varphi \notin \hat{G}'(E'')$. In $\mathsf{H}_L\mathcal{P}$ the subobject $(E' \Rightarrow E'') \vee (E'' \Rightarrow E')$ of $y(m)$ is the top element, so id_m lies in $\hat{G}(E' \Rightarrow E'')$ or in $\hat{G}(E'' \Rightarrow E')$. In the first case $\varphi \in \hat{G}'(E' \Rightarrow E'')$, the homomorphism φ sending id_m to φ; with $\varphi \in \hat{G}'(E')$ this gives $\varphi \in \hat{G}'(E'')$, a contradiction. The second case is symmetric. □

By Lemma 2.2 the forcing at a model $\hat{G}$, in the Kripke model $\mathcal{K}$, involves the homomorphisms out of $\hat{G}$; Lemma 3.1 says that these are totally preordered by $\varphi \preceq \psi$ iff $\psi = \chi\varphi$ for a homomorphism χ.

4. EI-categories: the models are strings

From now on $\mathcal{X}$ is a finite EI-category and L proves linearity. We use the following consequences of the definition. A split monomorphism or epimorphism of $\mathcal{X}$ is invertible: if $\rho\gamma = \mathrm{id}$, then $\gamma\rho$ is an idempotent endomorphism, hence invertible, hence the identity. So idempotents split, and a composite $\delta\gamma$ is invertible only if γ and δ are. If there is a non-invertible arrow $x \to x'$, then x and x' are not isomorphic; so the isomorphism classes of objects are strictly partially ordered by the existence of a non-invertible arrow. The non-invertible arrows out of x form a subfunctor $N_x \leq y(x)$.

Lemma 4.1. *A homomorphism of models of $\mathsf{H}_L\mathcal{P}$ whose underlying arrow is invertible in $\mathcal{X}$ is an isomorphism of models.*

Proof. We show, for all homomorphisms $\chi\colon \hat{G}' \to \hat{G}''$ with invertible underlying arrow at once and all $U \leq A$, that $b \in \hat{G}'(U)$ if $A(\chi)(b) \in \hat{G}''(U)$, by induction on the construction of U. This is clear for the subobjects in $\mathcal{P}$ and compatible with finite meets and joins and f^*; for $\exists_f$ it holds because χ is bijective on every sort. Let $A(\chi)(b) \in \hat{G}''(\forall_f U)$, $f\colon B \to A$; by Lemma 2.2 we have to show $\beta \in \hat{G}'''(U)$ for a homomorphism $\lambda\colon \hat{G}' \to \hat{G}'''$ and β over $A(\lambda)(b)$. By Lemma 3.1 there is a homomorphism μ with $\mu\chi = \lambda$ or with $\mu\lambda = \chi$. In the first case $A(\lambda)(b) = A(\mu)A(\chi)(b) \in \hat{G}'''(\forall_f U)$ and $\beta \in \hat{G}'''(U)$. In the second case λ is a split monomorphism, hence invertible, and so is $\mu\colon \hat{G}''' \to \hat{G}''$; $B(\mu)(\beta)$ lies over $A(\chi)(b)$, so $B(\mu)(\beta) \in \hat{G}''(U)$, and $\beta \in \hat{G}'''(U)$ by the induction hypothesis for μ and U. The case of $U \Rightarrow V$ is the same. □

A *string* is a sequence $s = (m_0 \xrightarrow{a_1} m_1 \to \ldots \xrightarrow{a_r} m_r)$ of non-invertible arrows of $\mathcal{X}$, $r \geq 0$; its *length* is $r + 1$, its *final segments* are the strings $s_{\geq j} = (m_j \to \cdots \to m_r)$, and an isomorphism of strings is a family of isomorphisms of the objects commuting with the arrows. The length of a string is at most the number of isomorphism classes of objects of $\mathcal{X}$. The string s is a Kripke model over the chain $0 < \cdots < r$, with $M_j = m_j$; if $r + 1 \leq n_L$ let $\hat{G}_s$ be the model of $\mathsf{H}_L\mathcal{P}$ given by the evaluation at the node 0. The evaluation at the node j is $\hat{G}_{s_{\geq j}}$, and $a_j \cdots a_1\colon \hat{G}_s \to \hat{G}_{s_{\geq j}}$ is a homomorphism.

Proposition 4.2. *Let $\mathcal{X}$ be a finite EI-category, L an intermediate logic proving linearity and $n = n_L$.*

(a) *Every model of $\mathsf{H}_L\mathcal{P}$ is isomorphic to $\hat{G}_s$ for a string s of length at most n.*
(b) *The homomorphisms $\hat{G}_s \to \hat{G}_{s'}$ are the arrows $\iota_0 a_j \cdots a_1$, for $0 \leq j \leq r$ and ι an isomorphism of strings from $s_{\geq j}$ onto s'; the index j is determined by s'. In particular $\hat{G}_s \cong \hat{G}_{s'}$ if and only if s and s' are isomorphic strings.*

Proof. Step 1: the string of a model. Let $\hat{G}$ be a model over m_0. By Lemma 3.1 the homomorphisms out of $\hat{G}$ are totally preordered by $\preceq$. If $\varphi \preceq \psi \preceq \varphi$, say $\psi = \chi\varphi$ and $\varphi = \chi'\psi$ with $\chi\colon \hat{G}' \to \hat{G}''$, then $\chi'\chi$ is an endomorphism of m' and $\chi\chi'$ one of m''; so χ is invertible in $\mathcal{X}$, and an isomorphism of models by Lemma 4.1. If $\varphi \prec \psi = \chi\varphi$, then χ is not an isomorphism of models, hence not invertible in $\mathcal{X}$, and the isomorphism class of the codomain of ψ is strictly above that of φ. So there are finitely many classes; choose representatives $\mathrm{id} = \varphi_0 \prec \varphi_1 \prec \cdots \prec \varphi_r$, $\varphi_j\colon \hat{G} \to \hat{G}_j$, and homomorphisms $a_j\colon \hat{G}_{j-1} \to \hat{G}_j$ with $\varphi_j = a_j\varphi_{j-1}$. The a_j are not invertible, and $s := (m_0 \xrightarrow{a_1} \ldots \xrightarrow{a_r} m_r)$ is a string.

Step 2: every homomorphism λ out of $\hat{G}_j$ is $\chi a_{j'} \cdots a_{j+1}$ for some $j' \geq j$ and an isomorphism of models χ. The homomorphism $\lambda\varphi_j$ out of $\hat{G}$ is equivalent to some $\varphi_{j'}$, and $j' \geq j$ because $\varphi_j \preceq \lambda\varphi_j$; so $\lambda\varphi_j = \iota a_{j'} \cdots a_{j+1}\varphi_j$ with an isomorphism of models ι. By Lemma 3.1 at $\hat{G}_j$, λ and $\iota a_{j'} \cdots a_{j+1}$ differ by a homomorphism between their codomains in one of the two directions; it is an endomorphism in $\mathcal{X}$, hence invertible, hence an isomorphism of models.

Step 3: $\hat{G}_j(U)$ is the value of U at the node j of the Kripke model s, for all j and all $U \leq A$, where the forcing relation of s is defined by the usual clauses, whether or not the chain is a frame of L. By induction on the construction of U: both sides agree on the subobjects in $\mathcal{P}$ and are compatible with the coherent operations; for $\forall_f U$, by Lemma 2.2 and Step 2, $\alpha \in \hat{G}_j(\forall_f U)$ iff for all $j' \geq j$, all isomorphisms of models χ out of $\hat{G}_{j'}$ and all β over $A(\chi a_{j'} \cdots a_{j+1})(\alpha)$ we have β in the value of U at the codomain of χ, iff, applying χ^{-1}, for all $j' \geq j$ and all β over $A(a_{j'} \cdots a_{j+1})(\alpha)$ we have $\beta \in \hat{G}_{j'}(U)$, which by induction is the forcing clause at the node j. The case of $\Rightarrow$ is the same.

Step 4: $r+1 \leq n$. Let $U_i \leq y(m_0)$ be the image of $y(a_i \cdots a_1)$, for $1 \leq i \leq r$, the arrows out of m_0 which factor through $a_i \cdots a_1$. The arrow $a_j \cdots a_1$ lies in U_i iff $j \geq i$: if $a_j \cdots a_1 = \beta a_i \cdots a_1$ with $j < i$, then $\beta a_i \cdots a_{j+1}$ is an endomorphism of m_j, so a_{j+1} would be a split monomorphism. So at the element id_{m_0} the U_i are forced exactly at the nodes $j \geq i$, and every valuation of propositional variables by up-sets of the chain $0 < \cdots < r$ is obtained from the U_i, the top and the bottom. For a formula $\lambda(p_1, \ldots, p_k)$ of L and such a valuation, the subobject $\lambda(U_{i_1}, \ldots, U_{i_k})$ of $y(m_0)$ is the

top element in $\mathsf{H}_L\mathcal{P}$, so id_{m_0} lies in its value under $\hat{G}$; by Step 3 this value is computed by forcing in the chain, where at the element id_{m_0} it is the propositional forcing of λ. So λ is valid in the chain, the chain is a frame of L, and $r+1 \leq n$. Hence $\hat{G}_s$ is defined, and $\hat{G} \cong \hat{G}_s$ by Step 3; this is (a).

Step 5: (b). Let s be a string of length at most n. We show first that every homomorphism $\psi\colon \hat{G}_s \to \hat{G}'$ is $\chi a_i \cdots a_1$ with χ an isomorphism of models. Let i be maximal with $a_i \cdots a_1 \preceq \psi$, and $\psi = \lambda a_i \cdots a_1$ with $\lambda\colon \hat{G}_{s_{\geq i}} \to \hat{G}'$; it suffices to show that λ is invertible in $\mathcal{X}$. If $i = r$, the node r is the top of the chain, so the forcing there is classical and $\mathrm{id}_{m_r} \in \hat{G}_{s_{\geq r}}(\neg N_{m_r})$; then $\lambda \in \hat{G}'(\neg N_{m_r})$, so $\lambda \notin N_{m_r}(m')$. Let $i < r$ and suppose that λ is not invertible. By Lemma 3.1 and the maximality of i there is a homomorphism μ with $\mu\lambda = a_{i+1}$. At the node i of s the element id_{m_i} forces $N_{m_i} \Rightarrow I$, where $I \leq y(m_i)$ is the image of $y(a_{i+1})$: at the nodes $j > i$ it has become $a_j \cdots a_{i+1}$. So $\lambda \in \hat{G}'(N_{m_i} \Rightarrow I)$, and as $\lambda \in N_{m_i}$, $\lambda = \beta a_{i+1}$ for an arrow β of $\mathcal{X}$. Then $a_{i+1} = \mu\beta a_{i+1}$, the endomorphism $\mu\beta$ is invertible, so β and μ are invertible, μ is an isomorphism of models and $a_{i+1} \preceq \lambda$, contradicting the maximality of i.

It remains to show that the isomorphisms of models $\chi_0\colon \hat{G}_u \to \hat{G}_t$ are the first components of the isomorphisms of strings $u \to t$. An isomorphism of strings is an isomorphism of Kripke models and preserves forcing. Conversely let χ_0 be an isomorphism of models, $u = (m_0 \xrightarrow{a_1} \cdots)$ and $t = (m_0' \xrightarrow{b_1} \cdots)$. It induces a bijection between the homomorphisms out of $\hat{G}_u$ and out of $\hat{G}_t$ which preserves $\preceq$; by what was shown, the classes are those of the $a_i \cdots a_1$ and of the $b_i \cdots b_1$, which are pairwise inequivalent since their codomains are pairwise non-isomorphic. So u and t have the same length, and $b_1\chi_0 = \chi_1 a_1$ for an isomorphism of models $\chi_1\colon \hat{G}_{u_{\geq 1}} \to \hat{G}_{t_{\geq 1}}$; by induction on the length χ_1 extends to an isomorphism of strings. The index j in (b) is determined because the codomains of the $a_j \cdots a_1$ are pairwise non-isomorphic. □

Definition 4.3. $\mathcal{S}_n = \mathcal{S}_n(\mathcal{X})$ is the category whose objects are the strings of length at most n, one in each isomorphism class, and whose arrows $s \to s'$ are the arrows $\iota_0 a_j \cdots a_1\colon m_0 \to m_0'$ of $\mathcal{X}$ of Proposition 4.2(b); $\pi\colon \mathcal{S}_n \to \mathcal{X}$ is the first object, a faithful functor. We write $\mathrm{Aut}_{\mathcal{S}}(s) \leq \mathrm{Aut}_{\mathcal{X}}(m_0)$ for the group of the first components of the automorphisms of the string s.

By Proposition 4.2 the category $\mathcal{K}$ of models of $\mathsf{H}_L\mathcal{P}$ is $\mathcal{S}_n$, a finite category. For a poset $\mathcal{X}$ the strings are the strict chains and $\mathcal{S}_n$ is the poset of the strict chains with at most n elements, ordered by final segments, of [6, §6]; this description of the propositional case by chains is known, from the free Gödel algebras over finite distributive lattices [1] and over bounded distributive lattices in the subvarieties [2].

5. The pretopos completion of the translated theory

Lemma 5.1. *Let A be in $\mathcal{P}$ and S a subfunctor of $\pi^*A\colon \mathcal{S}_n \to$ **FinSet**. Then there is a unique $V \leq A$ in $\mathsf{H}_L\mathcal{P}$ with $\hat{G}_s(V) = S(s)$ for all s.*

Proof. Uniqueness is the conservativity of the evaluation. For a string s with first object m, a string t and an arrow $\varphi\colon m \to t_0$ which is not an arrow $s \to t$ of $\mathcal{S}_n$ there is, by Lemma 2.1, $U \leq y(m)$ with $\mathrm{id}_m \in \hat{G}_s(U)$ and $\varphi \notin \hat{G}_t(U)$. Let $\tau_s \leq y(m)$ be the meet of these finitely many U. Then $\varphi \in \hat{G}_t(\tau_s)$ if and only if φ is an arrow $s \to t$ of $\mathcal{S}_n$. Let V be the join of the $\exists_{q_a}\tau_s$ over the strings s and the $a \in S(s)$. Then $\hat{G}_t(\exists_{q_a}\tau_s)$ is the set of the $A(\varphi)(a)$, $\varphi\colon s \to t$ in $\mathcal{S}_n$, which is contained in $S(t)$ and contains a if $t = s$. □

Lemma 5.2. *Let $F\colon \mathcal{H} \to \mathcal{E}$ be a coherent functor from a small coherent category to a pretopos which is conservative and full on subobjects, and such that every object of $\mathcal{E}$ is a quotient of a finite coproduct of objects $F(X)$. Then the coherent functor $\mathsf{P}\mathcal{H} \to \mathcal{E}$ extending F is an equivalence.*

Proof. An object of $\mathsf{P}\mathcal{H}$ is a quotient $(\coprod_i X_i)/R$ with R given by subobjects $R_{ij} \leq X_i \times X_j$ in $\mathcal{H}$, and a morphism to $(\coprod_j X'_j)/R'$ is a family of subobjects of the $X_i \times X'_j$ which is a functional relation compatible with R and R'; the same holds in $\mathcal{E}$ for the quotients of the coproducts of the $F(X_i)$. Since F is full on subobjects, every such family in $\mathcal{E}$ comes from $\mathcal{H}$, and since F is conservative, the conditions on the families, which are inequalities between subobjects built by the coherent operations, hold in $\mathcal{H}$ if they hold in $\mathcal{E}$, and two families are equal if their images are. So the extension is full and faithful, and an equivalence relation on $\coprod_i F(X_i)$ in $\mathcal{E}$ comes from one in $\mathsf{P}\mathcal{H}$; hence the extension is essentially surjective. □

Proof of Theorem 1.1. By Lemma 2.2 and Proposition 4.2 the evaluation $F\colon \mathsf{H}_L\mathcal{P} \to \mathbf{FinSet}^{\mathcal{S}_n}$, $F(U)(s) = \hat{G}_s(U)$, is a conservative Heyting functor with $F(\eta A) = \pi^*A$. It is full on subobjects: a subfunctor of $F(U)$, $U \leq A$, is a subfunctor of π^*A, hence $F(V)$ for some $V \leq A$ by Lemma 5.1, and $V \leq U$ by conservativity. Every $W\colon \mathcal{S}_n \to$ **FinSet** is a quotient of a finite coproduct of representable functors $\mathcal{S}_n(s,-)$, and $\mathcal{S}_n(s,-)$ is a subfunctor of $\pi^*y(m_0)$, π being faithful. By Lemma 5.2, $\mathsf{PH}_L\mathcal{P} \simeq \mathbf{FinSet}^{\mathcal{S}_n}$, and $\mathsf{H}_L\mathcal{P}$ corresponds to the full subcategory of the subfunctors of the π^*A. So $\mathsf{H}_L\mathcal{P}$ is a pretopos, that is $c^L_{\mathcal{P}}$ is an equivalence, if and only if every W is isomorphic to a subfunctor of some π^*A. A morphism $W \to \pi^*Y$ factors through the unit $W \to \pi^*\mathrm{Lan}_\pi W$, and $\mathrm{Lan}_\pi W$ has finite values, $\mathcal{S}_n$ being finite; so W embeds in some π^*Y if and only if the unit is injective. □

For classical logic the models can be described without the assumption on the endomorphisms.

Proposition 5.3 (classical logic)**.** *Let $\mathcal{X}$ be a finite category in which idempotents split, $\mathcal{X}^\times$ the groupoid of its isomorphisms and L classical logic.*

Then the models of $\mathsf{H}_L\mathcal{P}$ are the objects of $\mathcal{X}$, the homomorphisms are the isomorphisms of $\mathcal{X}$, $\mathsf{PH}_L\mathcal{P} \simeq \mathbf{FinSet}^{\mathcal{X}^\times}$, and $c^L_{\mathcal{P}}$ is an equivalence.

Proof. The evaluation at an object m is a coherent functor $\mathcal{P} \to \mathbf{Set}$, and $\mathbf{Set}$ is Boolean; so it extends to a Heyting functor on $\mathsf{H}_L\mathcal{P}$, uniquely up to isomorphism [6, §2]. Since $\mathsf{H}_L\mathcal{P}$ is Boolean, every coherent functor on it is Heyting [5, Lemma 2.1]; so there is exactly one model over each m, and the isomorphisms of $\mathcal{X}$ are isomorphisms of models. Let $\varphi\colon m \to m'$ be a homomorphism of models. It preserves complements, hence reflects the subobjects of $\mathsf{H}_L\mathcal{P}$. For an object x of $\mathcal{X}$ and $k \geq 1$ let $E_k \leq 1$ be the image of the complement, in $y(x)^k$, of the union of the diagonals; it is true in the model over m iff $\mathcal{X}(x, m)$ has at least k elements. As φ preserves and reflects the E_k and reflects the diagonal of $y(x)$, the map $\mathcal{X}(x, m) \to \mathcal{X}(x, m')$ is an injection between sets of the same cardinality, for every x; so φ is invertible. Hence the category $\mathcal{K}$ of models is a skeleton of $\mathcal{X}^\times$. The proofs of Lemma 5.1 and of the equivalence $\mathsf{PH}_L\mathcal{P} \simeq \mathbf{FinSet}^{\mathcal{K}}$ in Theorem 1.1 used only Lemma 2.1, the conservativity of the evaluation, the finiteness of $\mathcal{K}$ and the faithfulness of $\pi\colon \mathcal{K} \to \mathcal{X}$, so they apply. Finally let $W\colon \mathcal{K} \to \mathbf{FinSet}$ and m in $\mathcal{K}$; $\mathrm{Lan}_\pi W(m)$ is the colimit of $W(m')$ over the category of the pairs $(m', g\colon m' \to m)$, whose arrows $(m', g'v) \to (m'', g')$ are the isomorphisms v. In it $g'v$ is invertible iff g' is, so the pairs with g invertible form a union of connected components, with the terminal object (m, id), and the unit $W(m) \to \mathrm{Lan}_\pi W(m)$ is injective. □

6. When the translation commutes with the imaginaries

For a string s with first object m let $\mathcal{J}_s$ be the category of the pairs (t, g), t a string and $g\colon t_0 \to m$ an arrow of $\mathcal{X}$, an arrow $(t, g) \to (t', g')$ being an arrow $v\colon t \to t'$ of $\mathcal{S}_n$ with $g'v = g$. Then $\mathrm{Lan}_\pi W(m)$ is the colimit of $W(t)$ over $\mathcal{J}_s$, and the unit at s is the map from $W(s)$, at the object (s, id). Let $\mathcal{J}^0_s$ be the full subcategory of the (t, g) with g an arrow $t \to s$ of $\mathcal{S}_n$; it has the terminal object (s, id), so the colimit of W over $\mathcal{J}^0_s$ is $W(s)$. An arrow of $\mathcal{J}_s$ into an object of $\mathcal{J}^0_s$ has its domain in $\mathcal{J}^0_s$. Let $\mathcal{J}^c_s \supseteq \mathcal{J}^0_s$ be the connected component of (s, id) in $\mathcal{J}_s$, and $C_s(t)$ the set of the arrows g with (t, g) in $\mathcal{J}^c_s$; so $\mathcal{S}_n(t, s) \subseteq C_s(t) \subseteq \mathcal{X}(t_0, s_0)$, and $gv \in C_s(t')$ iff $g \in C_s(t)$, for $v\colon t' \to t$ in $\mathcal{S}_n$.

Proposition 6.1. *Suppose that for all arrows $v\colon t' \to t$ and $u\colon t' \to s$ of $\mathcal{S}_n$, every arrow $g\colon t_0 \to s_0$ of $\mathcal{X}$ with $gv = u$ is an arrow $t \to s$ of $\mathcal{S}_n$. Then $c^L_{\mathcal{P}}$ is an equivalence.*

Proof. The hypothesis says that an arrow of $\mathcal{J}_s$ out of an object of $\mathcal{J}^0_s$ has its codomain in $\mathcal{J}^0_s$. So $\mathcal{J}^0_s$ is a union of connected components of $\mathcal{J}_s$, the colimit of W over $\mathcal{J}_s$ is the disjoint union of $W(s)$ and the colimit over the other components, and the unit is injective. □

Corollary 6.2. *Let X be a finite poset and Γ a finite group, regarded as a category with one object. For $\mathcal{X} = X \times \Gamma$, in particular for $\mathcal{X} = X$, the comparison $c^L_{\mathcal{P}}$ is an equivalence for every L proving linearity.*

Proof. An arrow of $\mathcal{X}$ is a pair $(x \leq x', \gamma)$; it is invertible iff $x = x'$, so $\mathcal{X}$ is an EI-category, and a string is a strict chain $x_0 < \cdots < x_r$ with elements $\gamma_1, \ldots, \gamma_r$. For strings t, s whose chains are $x_0 < \cdots < x_r$ and $x_j < \cdots < x_r$, every $\gamma_0 \in \Gamma$ is the first component of a unique isomorphism of strings from $t_{\geq j}$ onto s, the other components being determined successively. So the arrows $t \to s$ of $\mathcal{S}_n$ are all the arrows $t_0 \to s_0$ of $\mathcal{X}$ if the chain of s is a final segment of that of t, and there are none otherwise. Let $v\colon t' \to t$, $u\colon t' \to s$ and $g\colon t_0 \to s_0$ with $gv = u$. The chains of t and s are final segments of that of t', and the first element of t is below that of s; so the chain of s is a final segment of that of t, and g is an arrow of $\mathcal{S}_n$. Proposition 6.1 applies. □

For $\Gamma = 1$ Corollary 6.2 is the theorem of [6] that the translation into a logic proving linearity commutes with the imaginaries of the finite propositional theories. For $X = \{0 < 1\}$ and $\Gamma = \mathbb{Z}/2$ the pretopos $\mathcal{P}$ is that of [5, Ex. 7.16], a proposition p and a sort with exactly two elements and decidable equality; over intuitionistic logic a twisted form of this sort is not in the image of the comparison, and it needs a Kripke model over a fork.

Corollary 6.3. *If $n_L \leq 2$, in particular for classical logic and for the logic of the two-element chain, $c^L_{\mathcal{P}}$ is an equivalence for every finite EI-category $\mathcal{X}$.*

Proof. Let $v\colon t' \to t$ and $u\colon t' \to s$ be arrows of $\mathcal{S}_n$ and $gv = u$. If v is invertible, $g = uv^{-1}$ is an arrow of $\mathcal{S}_n$. Otherwise $t' = (m_0 \xrightarrow{a} m_1)$, t has length one and $v = \iota a$ with $\iota\colon m_1 \to t_0$ invertible. If u were invertible, a would be a split monomorphism; so $u = \iota' a$ with $\iota'\colon m_1 \to s_0$ invertible, and s has length one. Then $\iota'^{-1} g \iota$ is an endomorphism of m_1, so g is invertible, and an invertible arrow between strings of length one is an arrow of $\mathcal{S}_n$. Proposition 6.1 applies. □

The hypothesis of Proposition 6.1 says that $C_s(t) = \mathcal{S}_n(t, s)$. It is the simplest case of the following.

Proposition 6.4 (retraction)**.** *Suppose that for every string s of length at most n there are maps $r_t\colon C_s(t) \to \mathcal{S}_n(t, s)$, t in $\mathcal{S}_n$, with $r_t(\kappa) = \kappa$ for $\kappa \in \mathcal{S}_n(t, s)$ and $r_{t'}(gv) = r_t(g)v$ for $g \in C_s(t)$ and $v\colon t' \to t$ in $\mathcal{S}_n$. Then $c^L_{\mathcal{P}}$ is an equivalence.*

Proof. The colimit of W over $\mathcal{J}_s$ is the disjoint union of the colimits over the connected components. On the colimit over $\mathcal{J}_s^c$ let ρ send the class of $w \in W(t)$ at the object (t, g) to $W(r_t(g))(w) \in W(s)$. This is well defined: for $v\colon (t', gv) \to (t, g)$ and $w' \in W(t')$ the elements w' at (t', gv) and $W(v)w'$ at (t, g) are both sent to $W(r_t(g)v)(w')$. Since $r_s(\mathrm{id}) = \mathrm{id}$, ρ is a left inverse of the map from $W(s)$ to the colimit over $\mathcal{J}_s^c$, which is a summand of $\mathrm{Lan}_\pi W(s_0)$; so the unit is injective. □

Corollary 6.5. *If every automorphism of $\mathcal{X}$ is an identity, then $c^L_{\mathcal{P}}$ is an equivalence for every L proving linearity.*

Proof. Two arrows $t \to s$ of $\mathcal{S}_n$ are $\iota_0 p$ and $\iota'_0 p$ for the same composite p of arrows of t and isomorphisms of strings ι, ι' onto s; these differ by an automorphism of the string s, whose first component is the identity by assumption. So $\mathcal{S}_n(t,s)$ has at most one element. We show that it is non-empty if $C_s(t)$ is; then the unique maps $r_t\colon C_s(t) \to \mathcal{S}_n(t,s)$ satisfy the conditions of Proposition 6.4. It suffices to show, for $v\colon t' \to t$ in $\mathcal{S}_n$ and $g\colon t_0 \to s_0$ in $\mathcal{X}$ with $\mathcal{S}_n(t',s) \neq \emptyset$, that $\mathcal{S}_n(t,s) \neq \emptyset$, the converse being clear. Up to isomorphism $t = t'_{\geq j}$ and $s = t'_{\geq k}$, with $t' = (m_0 \to \cdots \to m_r)$. If $k < j$, the composite $p\colon m_k \to m_j$ of the arrows of t' is not invertible, and gp is an endomorphism of m_k, so p would be a split monomorphism. So $k \geq j$, and s is a final segment of t. □

Example 6.6. Let $\mathcal{X}$ have the objects 0, 1, 2 and, besides the identities, the arrows $a\colon 0 \to 1$, $c, c'\colon 1 \to 2$ and $d = ca = c'a\colon 0 \to 2$:

$$0 \xrightarrow{\ a\ } 1 \underset{c'}{\overset{c}{\rightrightarrows}} 2$$

By Corollary 6.5, $c^L_{\mathcal{P}}$ is an equivalence for every L proving linearity. The hypothesis of Proposition 6.1 fails for $n \geq 3$: for $v = a\colon (0 \xrightarrow{a} 1 \xrightarrow{c} 2) \to (1 \xrightarrow{c} 2)$ and $u = d\colon (0 \xrightarrow{a} 1 \xrightarrow{c} 2) \to (2)$ one has $c'v = u$, and c' is not an arrow $(1 \xrightarrow{c} 2) \to (2)$ of $\mathcal{S}_n$. The retraction sends c' to c.

Proposition 6.7 (subgroup obstruction)**.** *Let t be a string of length at most n with first object m, $G = \mathrm{Aut}_{\mathcal{X}}(m)$, $H = \mathrm{Aut}_{\mathcal{S}}(t) \leq G$, and let $v_i\colon t'_i \to t$, $i = 1, 2$, be arrows of $\mathcal{S}_n$ with stabilisers $S_i = \{g \in G \mid gv_i = v_i\}$. If*

$$\langle H \cap S_1,\ H \cap S_2 \rangle \ \neq\ H \cap \langle S_1, S_2 \rangle,$$

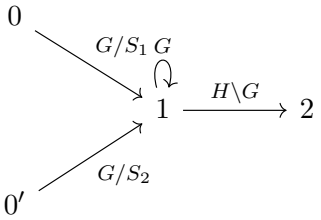
then $c^L_{\mathcal{P}}$ is not an equivalence.

Proof. Let W be the pushout in $\mathbf{FinSet}^{\mathcal{S}_n}$ of the morphisms $\mathcal{S}_n(t,-) \to \mathcal{S}_n(t'_i,-)$, $\kappa \mapsto \kappa v_i$, and $x_i \in W(t'_i)$ the class of the identity; so $W(v_1)x_1 = W(v_2)x_2 =: y$. The arrows $t'_i \to t$ of $\mathcal{S}_n$ are the hv_i, $h \in H$: by Proposition 4.2(b) the index of the final segment is determined, and two isomorphisms of strings onto t differ by an automorphism of t. So $W(t)$ is the quotient of $Hv_1 \sqcup Hv_2$ by the relation generated by $hv_1 \sim hv_2$, and $hv_i = h'v_i$ iff $h^{-1}h' \in S_i$; hence the stabiliser of y in H, for the action of H on $W(t)$ by the arrows $t \to t$, is $\langle H \cap S_1, H \cap S_2 \rangle$. Suppose that $\theta\colon W \to \pi^*Y$ is injective, and let $\bar{y} = \theta_t(y) \in Y(m)$. By naturality $\bar{y} = Y(v_i)\theta(x_i)$, so $Y(g)\bar{y} = \bar{y}$ for $g \in S_i$, hence for $g \in \langle S_1, S_2 \rangle$. For $h \in H \cap \langle S_1, S_2 \rangle$ we get $\theta_t(W(h)y) = Y(h)\bar{y} = \theta_t(y)$, so $W(h)y = y$ and $h \in \langle H \cap S_1, H \cap S_2 \rangle$. So under the hypothesis W is not a subfunctor of any π^*Y, and Theorem 1.1 applies. □

Example 6.8. Let $G = \mathbb{Z}/2 \times \mathbb{Z}/2$, $S_1 = \langle (1,0) \rangle$, $S_2 = \langle (0,1) \rangle$ and $H = \langle (1,1) \rangle$. Let $\mathcal{X}$ have the objects 0, 0′, 1, 2, with $\mathrm{Aut}(1) = G$ and trivial automorphism groups otherwise, and the arrows

$$\mathcal{X}(0,1) = G/S_1, \qquad \mathcal{X}(0',1) = G/S_2, \qquad \mathcal{X}(1,2) = H\backslash G,$$
$$\mathcal{X}(0,2) = \{d\}, \qquad \mathcal{X}(0',2) = \{d'\},$$

composition being given by the actions of G on the cosets and by the unique maps to the one-element sets:



This is a category with fifteen arrows, and an EI-category. Let $a = S_1 \in \mathcal{X}(0,1)$, $a' = S_2 \in \mathcal{X}(0',1)$ and $c = H \in \mathcal{X}(1,2)$. The string $t = (1 \xrightarrow{c} 2)$ has $\mathrm{Aut}_{\mathcal{S}}(t) = \{g \mid cg = c\} = H$, and the arrows $a\colon (0 \xrightarrow{a} 1 \xrightarrow{c} 2) \to t$ and $a'\colon (0' \xrightarrow{a'} 1 \xrightarrow{c} 2) \to t$ of $\mathcal{S}_3$ have the stabilisers S_1 and S_2. Here $H \cap S_1 = H \cap S_2 = 1$ and $\langle S_1, S_2 \rangle = G$; so by Proposition 6.7 the comparison $c^L_{\mathcal{P}}$ is not an equivalence for the logics L proving linearity with $n_L \geq 3$, in particular for the Gödel–Dummett logic. In terms of Kripke models: the functor W of the proof has two points at a node of type t, exchanged by $(1,1)$; a point of this kind which comes from a node below along a and from a node below along a' must, in an imaginary of $\mathcal{P}$, be fixed by $(1,0)$ and by $(0,1)$.

The same construction gives an obstruction which does not refer to automorphisms of the first object.

Proposition 6.9 (one-step obstruction)**.** *Let s, t be strings of length at most n, $g\colon t_0 \to s_0$ an arrow of $\mathcal{X}$, and $v_i\colon t_i \to t$, $i = 1,2$, arrows of $\mathcal{S}_n$ such that gv_1 and gv_2 are arrows $t_i \to s$ of $\mathcal{S}_n$. Let $\sim$ be the equivalence relation on $\mathcal{S}_n(t_1, s) \sqcup \mathcal{S}_n(t_2, s)$ generated by $\kappa v_1 \sim \kappa v_2$, $\kappa \in \mathcal{S}_n(t, s)$. If $gv_1 \not\sim gv_2$, then $c^L_{\mathcal{P}}$ is not an equivalence.*

Proof. Let W be the pushout of the morphisms $\mathcal{S}_n(t, -) \to \mathcal{S}_n(t_i, -)$, $\kappa \mapsto \kappa v_i$, and $x_i \in W(t_i)$ the classes of the identities, so that $W(v_1)x_1 = W(v_2)x_2 =: y$. Then $W(s)$ is the quotient of $\mathcal{S}_n(t_1, s) \sqcup \mathcal{S}_n(t_2, s)$ by $\sim$, and the elements $z_i := W(gv_i)x_i$, the classes of the gv_i, are different. For a morphism $\theta\colon W \to \pi^* Y$ we have $\theta_s(z_i) = Y(gv_i)\theta_{t_i}(x_i) = Y(g)Y(v_i)\theta_{t_i}(x_i) = Y(g)\theta_t(y)$ for $i = 1,2$; so θ is not injective, and Theorem 1.1 applies. □

If g is an arrow $t \to s$ of $\mathcal{S}_n$, then $gv_1 \sim gv_2$; so the hypothesis implies that (t, g) is an object of $\mathcal{J}^c_s$ outside $\mathcal{J}^0_s$, and that there is no retraction as in Proposition 6.4, which would give $gv_i = r_t(g)v_i$.

Example 6.10. Let $G = \mathbb{Z}/2 \times \mathbb{Z}/2$, $a = (1,0)$, $b = (0,1)$. Let $\mathcal{X}$ have the objects 0, $0'$, 1, 2, with $\mathrm{Aut}(2) = \langle b \rangle$ and trivial automorphism groups otherwise, and the arrows

$$\mathcal{X}(0,1) = \mathcal{X}(0,2) = G/\langle a \rangle, \qquad \mathcal{X}(1,2) = G,$$
$$\mathcal{X}(0',1) = \mathcal{X}(0',2) = G/\langle a+b \rangle,$$

composition being the addition of G. This is an EI-category with seventeen arrows. Let $s = (2)$, $t = (1 \xrightarrow{0} 2)$, so that $\mathcal{S}_3(t,s) = \{0, b\}$, and $g = a \in \mathcal{X}(1,2)$. Let $v_1 = \langle a \rangle \in \mathcal{X}(0,1)$ and $v_2 = \langle a+b \rangle \in \mathcal{X}(0',1)$, arrows of $\mathcal{S}_3$ into t from the strings $t_1 = (0 \xrightarrow{v_1} 1 \xrightarrow{0} 2)$ and $t_2 = (0' \xrightarrow{v_2} 1 \xrightarrow{0} 2)$. Then $gv_1 = a + \langle a \rangle = 0 \cdot v_1$ and $gv_2 = a + \langle a+b \rangle = b \cdot v_2$ are arrows $t_i \to s$ of $\mathcal{S}_3$. The relation $\sim$ is generated by $0 \cdot v_1 \sim 0 \cdot v_2$ and $b \cdot v_1 \sim b \cdot v_2$, and $b \cdot v_i \neq 0 \cdot v_i$; so $gv_1 \not\sim gv_2$, and by Proposition 6.9 the comparison is not an equivalence for $n_L \geq 3$. Proposition 6.7 does not apply to this category: the only object with a non-trivial automorphism is 2, the only string with first object 2 is (2), and for it $H = \mathrm{Aut}_{\mathcal{X}}(2)$, so that $H \cap S_i = S_i$.

The category of Example 6.10 is the first of a family in which the condition of Theorem 1.1 can be read off the lattice of subgroups of an abelian group.

Proposition 6.11 (a three-level family)**.** *Let G be a finite abelian group and $A, T_1, \dots, T_m$ subgroups of G. Let $\mathcal{X}$ have the objects $0_1, \dots, 0_m$, 1, 2, with $\mathrm{Aut}(2) = A$ and trivial automorphism groups otherwise, and the arrows*

$$\mathcal{X}(1,2) = G, \qquad \mathcal{X}(0_j,1) = \mathcal{X}(0_j,2) = G/T_j \quad (1 \leq j \leq m),$$

composition being the addition of G. Let $n_L \geq 3$. Then $c^L_{\mathcal{P}}$ is an equivalence if and only if

$$A \cap \sum_{j \in J} T_j = \sum_{j \in J} (A \cap T_j) \qquad \text{for every } J \subseteq \{1, \dots, m\}. \tag{1}$$

Proof. $\mathcal{X}$ is an EI-category, and its strings have length at most three, so that $\mathcal{S}_n = \mathcal{S}_3$. Choose representatives γ of the cosets of A in G, with 0 for A. Up to isomorphism the strings with last object 2 are $s = (2)$, $t_\gamma = (1 \xrightarrow{\gamma} 2)$, $u_{j,x,\gamma} = (0_j \xrightarrow{x} 1 \xrightarrow{\gamma} 2)$ with $x \in G/T_j$, and strings $p = (0_j \to 2)$. The arrows of $\mathcal{S}_3$ between them are, besides the identities and the automorphisms $a \in A$ of s: the arrows of $\mathcal{S}_3(t_\gamma, s) = A + \gamma \subseteq G$; the arrow $x \colon u_{j,x,\gamma} \to t_\gamma$; the arrows of $\mathcal{S}_3(u_{j,x,\gamma}, s) = A + \gamma + x \subseteq G/T_j$; and arrows $p \to s$. An arrow of $\mathcal{S}_3$ preserves the last object of a string.

Let s' be a string other than (2); its first object is 1 or some 0_j. If $v \colon t' \to t$ and $u \colon t' \to s'$ are arrows of $\mathcal{S}_3$ and $g \colon t_0 \to s'_0$ an arrow of $\mathcal{X}$ with $gv = u$, then $t_0 \neq 2$, and either $t_0 = 1$, in which case $s'_0 = 1$, the strings t and s' are the final segment of t' beginning at 1, and $g = \mathrm{id}$; or $t_0 = 0_j$, in which case $t = t'$, $v = \mathrm{id}$ and $g = u$. So the argument of Proposition 6.1 shows that the unit is injective at s'.

Let now $s = (2)$, and $W\colon \mathcal{S}_3 \to \mathbf{FinSet}$. For $v \in W(t_\gamma)$ let $J(v)$ be the set of the j such that $v = W(x)y$ for some $x \in G/T_j$ and $y \in W(u_{j,x,\gamma})$, let $T(v) = \sum_{j\in J(v)} T_j$, and $f_v\colon A + \gamma \to W(s)$, $f_v(\kappa) = W(\kappa)v$. Two remarks:

(a) $f_v(\kappa + a) = f_v(\kappa)$ for $a \in A \cap T_j$, $j \in J(v)$: if $v = W(x)y$, then $f_v(\kappa) = W(\kappa + x)y$, and $\kappa + a + x = \kappa + x$ in G/T_j;
(b) in the colimit of W over $\mathcal{J}_s$, the element v at (t_γ, g) equals v at $(t_\gamma, g + \tau)$, for $g \in G$, $\tau \in T_j$, $j \in J(v)$: both are equal to y at $(u_{j,x,\gamma}, g + x)$.

By (b), for $\kappa, \kappa' \in A + \gamma$ with $\kappa - \kappa' \in T(v)$ the unit sends $f_v(\kappa)$ and $f_v(\kappa')$ to the same element, since $f_v(\kappa)$ at (s, id) equals v at (t_γ, κ).

Necessity. Suppose that (1) fails for J, and let $\kappa \in A \cap \sum_{j\in J} T_j$ be outside $\sum_{j\in J}(A \cap T_j)$. Let $u_j = u_{j,0,0}$, and let W be the colimit of the morphisms $\mathcal{S}_3(t_0, -) \to \mathcal{S}_3(u_j, -)$, $j \in J$, given by composition with $0\colon u_j \to t_0$; so W is generated by elements $y_j \in W(u_j)$ with $W(0)y_j = v$ for all $j \in J$, v being the class of the identity of t_0. The set $\mathcal{S}_3(u_j, s)$ is the image of A in G/T_j, so $W(s)$ is the colimit of the quotient maps $A \to A/(A \cap T_j)$, $j \in J$, which is $A/\sum_{j\in J}(A \cap T_j)$, and f_v is the quotient map. So $f_v(\kappa) \neq f_v(0)$; since $J \subseteq J(v)$, these elements have the same image under the unit.

Sufficiency. Assume (1), and that $W(s)$ is not empty, with an element w_0. Let $\mathcal{J}^*$ be the full subcategory of $\mathcal{J}_s$ of the objects (t, g) with t one of s, t_γ, $u_{j,x,\gamma}$, or with $t = p$ and $g \in \mathcal{S}_3(p, s)$; it is a union of connected components of $\mathcal{J}_s$: an arrow of $\mathcal{S}_3$ preserves the last object, and an object (p, c) with $c \notin \mathcal{S}_3(p, s)$ has no arrows to or from other objects, as $c = av$ with $a \in A$ and $v \in \mathcal{S}_3(p, s)$ is in $\mathcal{S}_3(p, s)$. We define a map ρ from the colimit of W over $\mathcal{J}^*$ to $W(s)$. At (s, a) and at (p, c) let $\rho(w) = W(a)w$ and $\rho(w) = W(c)w$. At (t_γ, g) let $\rho(v) = f_v(\kappa)$ if there is $\kappa \in (g + T(v)) \cap (A + \gamma)$, and $\rho(v) = w_0$ otherwise; two such κ differ by an element of $A \cap T(v)$, which by (1) is the sum of the $A \cap T_j$, $j \in J(v)$, so by (a) this does not depend on κ. At $(u_{j,x,\gamma}, c)$ let $\rho(y)$ be the value of ρ on $W(x)y$ at (t_γ, g), for $g \in G$ with $g + x = c$; since g is determined modulo T_j and $j \in J(W(x)y)$, this does not depend on g. The map ρ is compatible with the arrows of $\mathcal{J}^*$: with $x\colon (u_{j,x,\gamma}, g + x) \to (t_\gamma, g)$ by definition; with $\kappa\colon (t_\gamma, a + \kappa) \to (s, a)$ since $\rho(v) = f_v(a + \kappa) = W(a)W(\kappa)v$ at $(t_\gamma, a + \kappa)$; with $\kappa + x\colon (u_{j,x,\gamma}, a + \kappa + x) \to (s, a)$, $\kappa \in A + \gamma$, since $\rho(y) = f_{W(x)y}(a + \kappa) = W(a)W(\kappa + x)y$; and with the arrows into (s, a) from the objects (s, a') and (p, c) by the functoriality of W. Since $\rho(w) = w$ at (s, id), the unit is injective at s. □

Example 6.12. (a) Example 6.10 is the case $G = \mathbb{Z}/2 \times \mathbb{Z}/2$, $A = \langle b \rangle$, $T_1 = \langle a \rangle$, $T_2 = \langle a + b \rangle$: here $A \cap (T_1 + T_2) = A$ and $A \cap T_j = 0$.

(b) *The retractions of Proposition 6.4 are not necessary.* Let $G = (\mathbb{Z}/2)^3$ with basis e_1, e_2, e_3, $A = \langle e_1, e_2 \rangle$ and

$$T_1 = \langle e_1, e_3 \rangle, \qquad T_2 = \langle e_2, e_1 + e_3 \rangle, \qquad T_3 = \langle e_1 + e_2, e_3 \rangle,$$

a category with 28 arrows. The groups $A \cap T_j$ are $\langle e_1 \rangle$, $\langle e_2 \rangle$, $\langle e_1 + e_2 \rangle$, and any two of them generate A; so (1) holds and $c^L_{\mathcal{P}}$ is an equivalence for every

L proving linearity. For $s = (2)$ and $t = (1 \xrightarrow{0} 2)$ the arrow $e_3 \colon 1 \to 2$ is in $C_s(t)$ and not in $\mathcal{S}_3(t,s) = A$: for the arrow $x \colon (0_j \xrightarrow{x} 1 \xrightarrow{0} 2) \to t$ the composite $e_3 + x$ is an arrow into s of $\mathcal{S}_3$, as $A + T_j = G$. A retraction would send e_3 to $\kappa \in A$ with $\kappa + x = e_3 + x$ in G/T_j for $j = 1, 2, 3$, that is with κ in each of the sets $(e_3 + T_j) \cap A$, which are $\{0, e_1\}$, $\{e_1, e_1 + e_2\}$ and $\{0, e_1 + e_2\}$; these meet in pairs and have no common element. So for $n \geq 3$ there is no retraction as in Proposition 6.4.

(c) *The obstruction of Proposition 6.9 is not the only one.* Let $m \geq 3$, $G = (\mathbb{Z}/2)^m$ with basis $e_1, \dots, e_m$, $T_j = \langle e_j \rangle$ and $A = \langle e_1 + \dots + e_m \rangle$; for $m = 3$ this is a category with 38 arrows. Then $A \cap \sum_{j \in J} T_j$ is A for $J = \{1, \dots, m\}$ and 0 otherwise, and $A \cap T_j = 0$; so $c^L_{\mathcal{P}}$ is not an equivalence for $n_L \geq 3$, while it is an equivalence for the full subcategory obtained by omitting any one of the objects 0_j. The hypothesis of Proposition 6.9 is never satisfied: it requires (t, g) in $\mathcal{J}^c_s$ outside $\mathcal{J}^0_s$, so, by the proof of Proposition 6.11, $s = (2)$ and, an object $u_{j,x,\gamma}$ receiving no arrow but the identity, $t = t_\gamma$, $g \in G$ outside $A + \gamma$, and, gv_i being an arrow of $\mathcal{S}_3$, $v_i \neq \mathrm{id}$, that is $v_i = x_i \colon u_{j_i,x_i,\gamma} \to t_\gamma$, with $g \in A + \gamma + T_{j_i}$ as gv_i is an arrow of $\mathcal{S}_3$. If $j_1 \neq j_2$, then $g - \gamma \in (A + T_{j_1}) \cap (A + T_{j_2}) = A$, which is excluded. If $j_1 = j_2$, then $g = \kappa + \tau$ with $\kappa \in A + \gamma$ and $\tau \in T_{j_1}$, so that $gv_i = \kappa v_i$ and $gv_1 \sim gv_2$.

In general the condition of Theorem 1.1 has to be tested on finitely many functors only.

Proposition 6.13 (a finite test)**.** *For a string t of length at most n let $I_t \colon \mathcal{S}_n \to$ **FinSet** be the functor sending t' to the set of the subsets of $\mathcal{S}_n(t', t)$, and $k \colon t' \to t''$ to the map $U \mapsto \{l \in \mathcal{S}_n(t'', t) \mid lk \in U\}$. Then $c^L_{\mathcal{P}}$ is an equivalence if and only if the unit $I_t \to \pi^* \mathrm{Lan}_\pi I_t$ is injective for every t. In particular it is decidable, for a finite EI-category $\mathcal{X}$ and $n \in \{1, 2, \dots, \infty\}$, whether $c^L_{\mathcal{P}}$ is an equivalence for the logics L with $n_L = n$.*

Proof. For $W \colon \mathcal{S}_n \to$ **FinSet** the morphisms $\theta \colon W \to I_t$ correspond to the subsets B of $W(t)$, by $B = \{w \mid \mathrm{id}_t \in \theta_t(w)\}$ and $\theta_{t'}(w) = \{l \mid W(l)w \in B\}$. For $w \neq w'$ in $W(t)$ the morphism given by $B = \{w\}$ separates w and w'; so W is a subfunctor of a finite product of functors I_t. The unit of W is injective if and only if W is a subfunctor of $\pi^* Y$ for some $Y \colon \mathcal{X} \to$ **FinSet**, every morphism $W \to \pi^* Y$ factoring through the unit; and the functors with this property are closed under subfunctors and, π^* preserving products, under finite products. So the units of all W are injective as soon as those of the I_t are, and Theorem 1.1 applies. The sets $I_t(t')$ and the values of $\mathrm{Lan}_\pi I_t$ are finite and computable, $\mathcal{S}_n$ being a finite category. □

7. Questions

Question 7.1. Give a closed description of the finite EI-categories $\mathcal{X}$ for which every finite functor on $\mathcal{S}_n(\mathcal{X})$ embeds in one pulled back from $\mathcal{X}$. By Proposition 6.13 the condition is decidable; by Example 6.12 the retractions

of Proposition 6.4 are not necessary for it, and the obstructions of Propositions 6.7 and 6.9 are not the only ones. Is there a description in terms of the lattices of subgroups of the automorphism groups of $\mathcal{X}$, of which Proposition 6.7 and the equalities (1) are instances?

Question 7.2. Describe the models of $\mathsf{H}_L\mathcal{P}$ when $\mathcal{X}$ is a finite category with split idempotents which is not an EI-category. Lemma 3.1 still holds, but a non-invertible endomorphism has a power which is a non-identity idempotent, whose splitting is a pair of non-invertible arrows with identity composite, so that there are strings of unbounded length. A related example outside the present setting, as it has infinite models: for the theory of pure equality the comparison is an equivalence only for classical logic [6].

Question 7.3. For the logics which prove the weak excluded middle and not linearity the comparison fails already for the four-element chain [6]. Is there a description of the models of the translated theory over a finite poset, in which the chains are replaced by the finite rooted frames of the logic?

Declarations

The author has no competing interests to declare.

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